

THARUN DERANGULA

Software Engineer | Applied AI & Full-Stack | Distributed Systems

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SUMMARY

Software engineer with 5 years building production AI and distributed systems with Python, Java, TypeScript, React, and cloud platforms. Shipped LLM and RAG experiences for Walmart's Sparky AI shopping agent, including vector retrieval and evaluation pipelines operating across millions of daily events.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript (ES6+), Go, SQL

Frontend: React, Next.js (SSR), Redux, Tailwind CSS, Jest, React Testing Library, Cypress

Backend: FastAPI, Spring Boot, Node.js, Express.js, REST APIs, GraphQL, gRPC, Microservices, Kafka, Redis, OpenAI API, Anthropic API

AI/ML: Generative AI, LLM Integration & Evaluation, RAG, Embeddings, pgvector, Vector Search, LangChain, AI Agents, Prompt Engineering

Cloud & DevOps: GCP, AWS, Docker, Kubernetes, Terraform, Helm, GitHub Actions, CI/CD, Datadog, Prometheus, Grafana

Databases: PostgreSQL, MongoDB, Redis, Snowflake, Elasticsearch, BigQuery, DynamoDB

Testing & Quality: pytest, JUnit, Jest, React Testing Library, Cypress, TDD, A/B Testing

Practices & Tools: System Design, Event-Driven Architecture, MLOps, Agentic Workflows, Agile/Scrum, Claude Code, GitHub Copilot, Codex

EXPERIENCE

Software Engineer, Walmart Global Tech — Sunnyvale, CA

April 2025 – Present

- Shipped customer-facing conversational shopping and cart features for Sparky, Walmart's AI shopping agent, across web and mobile with **React**, **Next.js**, and **TypeScript**, partnering with product, design, and ML teams from experimentation through launch.
- Improved AI recommendation relevance by 20% in A/B testing and lifted add-to-cart conversion by building **Python RAG** evaluation pipelines, integrating Walmart's Wallaby **LLM** through the Element ML Platform, and optimizing **pgvector** retrieval.
- Built **Python** and **FastAPI** services for prompt orchestration, embedding generation, and retrieval-quality evaluation, with **pytest** regression suites and **GitHub Actions** quality gates that caught **RAG** failures before production releases.
- Architected cart auto-build and intent-routing microservices in **Java**, **Spring Boot**, **Kafka**, and **PostgreSQL** processing millions of daily shopping events, with idempotent retries, dead-letter queues, and near real-time **GCP Pub/Sub** pipelines for inventory and fulfillment.
- Cut p95 latency by 35% on high-traffic recommendation endpoints through **Redis** caching, connection pooling, and **PostgreSQL** query-plan and index tuning, sustaining sub-50ms median latency during seasonal and Rollback traffic spikes.
- Enabled zero-downtime releases and cut deployment effort by 50% through **Docker**, **Kubernetes**, **Terraform**, Walmart Cloud-Native Platform (**WCNP**), and **GitHub Actions**, backed by **Datadog** and **Grafana** observability.
- Secured shopping and cart APIs with **JWT**, **OAuth 2.0**, and role-based access control (**RBAC**), while maintaining 85%+ automated test coverage across **Python**, **Java**, and **TypeScript** components.
- Authored technical design documents and led architecture reviews for the cart AI agent, auto-build, intent-routing, and **RAG** evaluation services, aligning product, ML, and platform teams on scalable delivery.
- Shortened delivery cycles by building AI assisted workflows with **Claude Code**, **GitHub Copilot**, and **GitHub Actions** to automate code generation, refactoring, test creation, and CI/CD validation across frontend and backend codebases.

Software Engineer, Kennesaw State University — Kennesaw, GA

January 2024 – December 2024

- Built KSU's Campus Hub with **React**, **Next.js**, and **TypeScript**, consolidating Owl Express, D2L Brightspace, DegreeWorks, and KSUMail into one responsive platform serving 40,000+ students and faculty.
- Improved dashboard load time by 30% by developing **Node.js** and **Spring Boot REST** APIs with **Redis** caching and query tuning across Ellucian Banner and Degree Works integrations.
- Reduced manual data correction by 40% by building **Java** and **Spring Boot** services over **PostgreSQL** with **REST**-webhook reconciliation for enrollment, grades, and financial aid data.
- Shipped low-disruption releases during peak registration by deploying on **AWS EC2** and **S3** with **Docker** and **GitHub Actions CI/CD**, cutting deployment cycle time by half.
- Caught regressions earlier by maintaining 80%+ coverage with **Jest** and **React Testing Library** and collaborating with UITS teams in Agile sprints through design and code reviews.

Software Engineer, HCL Tech — India

May 2020 – December 2022

- Built invoice tracking and payment workflows serving enterprise banking customers across regions, with **React** and **TypeScript** frontends backed by **Java** and **Spring Boot** APIs.
- Improved peak-period API response times via **Redis** caching, connection pooling, and **PostgreSQL** query tuning across backend services.
- Engineered real-time streaming data pipelines with **Kafka**, using idempotent consumers and dead-letter queues to guarantee reliable transaction processing during high-volume settlement windows.
- Cut batch processing time by 45% by building **ETL pipelines** with **Python**, **pandas**, **SQL**, and **GCP BigQuery** that ingested transaction data into partitioned warehouse tables for finance and compliance teams.
- Reduced data quality incidents by 35% by implementing automated validation and reconciliation checks across **Airflow** orchestrated workflows, eliminating manual correction work.
- Took releases from days to hours and cut production defects by 40% by modernizing legacy backends into microservices on **Docker**, **Terraform**, **GCP Cloud Run**, and **GKE** with **GitHub Actions CI/CD** and TDD (**JUnit**, **pytest**).

EDUCATION

Master of Science, Information Technology | Kennesaw State University

Bachelor of Science, Computer Science & Engineering | Lovely Professional University

Certifications: Google Professional Cloud Architect | AWS Certified Solutions Architect | Microsoft Generative AI Engineering Professional